

WEEK 7 – POWER BI REPORT

STUDENT PERFORMANCE ANALYSIS – DATA PREPARATION

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Department: Data Science and Artificial Intelligence

Software Used: Microsoft Power BI

Dataset: Student Performance in Exams

1. Overview

In Week 7, the Student Performance dataset was prepared for academic analysis using **Microsoft Power BI**. The main focus of this work was to improve the quality of the raw data by performing data cleaning, transformation, calculated column creation, and data restructuring.

The prepared dataset will be useful for developing a Student Performance Dashboard in the next stage of the project.

2. Aim of the Project

The main aim of this project is to transform the raw student examination data into a clean and analysis-ready dataset.

Key Goals

- Import the student dataset into Power BI.
 - Examine the structure and quality of the data.
 - Clean unwanted and inconsistent data.
 - Transform categorical and numerical fields.
 - Calculate total and average marks.
 - Categorize students according to performance.
 - Create Pass/Fail classification.
 - Restructure subject-wise data.
 - Create DAX measures for analysis.
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3. Dataset Description

The dataset contains information related to students and their examination performance.

Major Fields

Category	Fields
Student Details	Gender, Race/Ethnicity
Background	Parental Education, Lunch
Preparation	Test Preparation Course
Academic Scores	Math, Reading, Writing

These fields allow student performance to be examined from different perspectives.

Table.AddColumn(#"Changed Type2", "Grade", each if [Average] >= 80 then "A" else if [Average] >= 60 then "B" else if [Average] >= 40

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed
23	Male	Group D	Some College	Standard	None

Query Settings

PROPERTIES

Name
StudentsPerformance

All Properties

APPLIED STEPS

Source

Promoted Headers

Changed Type

Capitalized Each Word

Changed Type1

Cleaned Text

Trimmed Text

Added Index

Renamed Columns

Reordered Columns

Added Custom

Added Custom1

Changed Type2

Added Conditional Column

4. Power Query Data Preparation

The dataset was opened in **Power Query Editor** after importing it into Power BI.

The data was examined for:

- Incorrect data types
- Missing values
- Duplicate entries
- Invalid values
- Inconsistent categories
- Unnecessary columns

The identified issues were handled before proceeding with further transformations.

5. Data Cleaning Process

The following cleaning operations were carried out:

Duplicate Removal

Duplicate records were checked and removed where necessary.

Missing Value Handling

Missing or invalid values were reviewed and handled appropriately.

Data Type Correction

Columns were assigned suitable data types so that calculations could be performed correctly.

Category Standardization

Categorical values were reviewed and standardized for consistent analysis.

Student ID	gender	race/ethnicity	parental level of education	lunch	test preparation course
1	Female	Group B	Bachelor's Degree	Standard	None
2	Female	Group C	Some College	Standard	Completed
3	Female	Group B	Master's Degree	Standard	None
4	Male	Group A	Associate's Degree	Free/Reduced	None
5	Male	Group C	Some College	Standard	None
6	Female	Group B	Associate's Degree	Standard	None
7	Female	Group B	Some College	Standard	Completed
8	Male	Group B	Some College	Free/Reduced	None
9	Male	Group D	High School	Free/Reduced	Completed
10	Female	Group B	High School	Free/Reduced	None
11	Male	Group C	Associate's Degree	Standard	None
12	Male	Group D	Associate's Degree	Standard	None
13	Female	Group B	High School	Standard	None
14	Male	Group A	Some College	Standard	Completed
15	Female	Group A	Master's Degree	Standard	None
16	Female	Group C	Some High School	Standard	None
17	Male	Group C	High School	Standard	None
18	Female	Group B	Some High School	Free/Reduced	None
19	Male	Group C	Master's Degree	Free/Reduced	Completed
20	Female	Group C	Associate's Degree	Free/Reduced	None
21	Male	Group D	High School	Standard	None
22	Female	Group B	Some College	Free/Reduced	Completed

Query Settings
PROPERTIES
Name: StudentsPerformance
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Capitalized Each Word
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Added Index
Renamed Columns
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Added Custom
Added Custom1
Changed Type2
Added Conditional Column

6. Student Score Calculation

New calculated fields were created to summarize student performance.

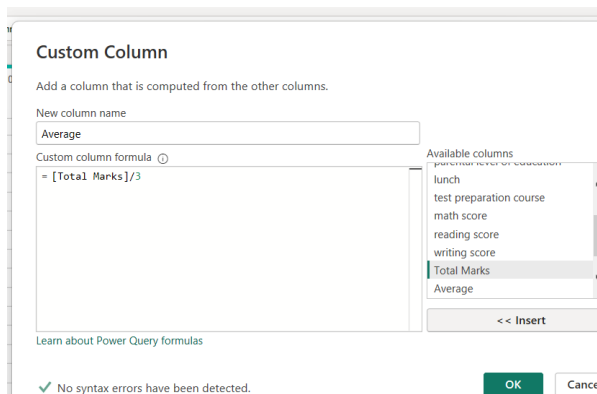
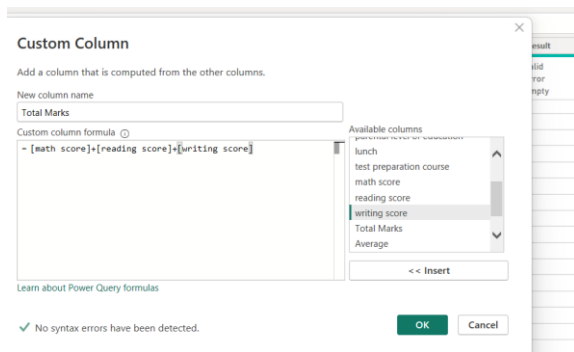
Total Marks

Total Score = Math Score + Reading Score + Writing Score

Average Marks

Average Score = Total Score ÷ 3

These values provide a simple way to measure the overall academic performance of each student.



7. Performance Categorization

Students were grouped into different performance levels using conditional logic.

Grade	Meaning
A	High Performance
B	Good Performance
C	Average Performance
F	Low Performance

This classification can be used later to study the distribution of student performance.

8. Pass and Fail Analysis

A conditional column was created to divide students into two groups:

Pass

Fail

The classification can later be used to calculate the overall pass percentage and compare performance between different student groups.

Table.AddColumn(AddedConditionalColumn, "Result", Each If [Average] >= 40 Then Pass Else Fail)

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
Result

	Column Name	Operator	Value		Output
If	Average	is greater than or...	40	Then	Pass

Add Clause

Else
Fail

OK Cancel

9. Subject Data Transformation

The three subject score columns were:

- Math Score
- Reading Score
- Writing Score

These columns were **unpivoted** in Power Query.

After transformation, the data was organized into:

New Field	Description
Subject	Name of the examination subject
Score	Score obtained by the student

This structure makes subject-wise comparison easier.

Queries [4] Table: RenameColumns(#"Unpivoted Columns",{"Attribute", "Subject"}, {"Value", "Score"})

Student ID	Subject	Score
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	1 math score	72
2	1 reading score	72
3	1 writing score	74
4	2 math score	69
5	2 reading score	90
6	2 writing score	88
7	3 math score	90
8	3 reading score	95
9	3 writing score	93
10	4 math score	47
11	4 reading score	57
12	4 writing score	44
13	5 math score	76
14	5 reading score	78
15	5 writing score	75
16	6 math score	71
17	6 reading score	83
18	6 writing score	78
19	7 math score	88
20	7 reading score	95
21	7 writing score	92
22	8 math score	40
23	8 reading score	43
24	8 writing score	24

10. Data Model Preparation

After completing the transformations, the dataset was reviewed before loading it into Power BI.

The following steps were performed:

- Removed unnecessary fields.
- Checked the transformed table.
- Reviewed keys and relationships.
- Verified the structure of the final dataset.
- Prepared the table for visualization.

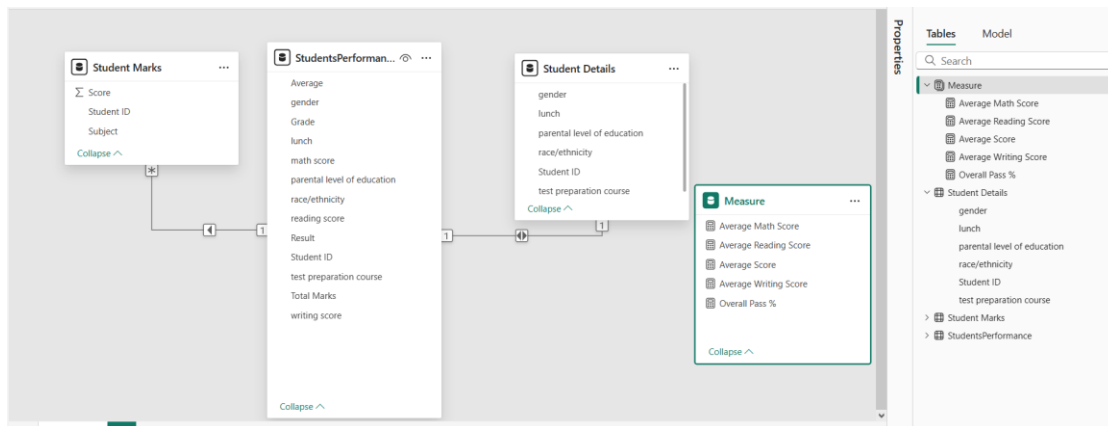
11. DAX Measures

Basic DAX measures were created to support the upcoming dashboard.

Measures Created

- Average Math Score
- Average Reading Score
- Average Writing Score
- Overall Average Score
- Overall Pass %

These measures provide summarized information for academic performance analysis.



12. Data Verification

The final dataset was checked to ensure that the transformations were working correctly.

The following were verified:

- Total Score calculation
- Average Score calculation
- Grade classification
- Pass/Fail classification
- Subject transformation
- DAX measure results

The transformed data was then considered ready for visualization.

13. Expected Analysis

The prepared dataset can be used to study:

- Overall student academic performance.
 - Subject-wise performance.
 - Differences between student groups.
 - Effect of test preparation on observed scores.
 - Relationship between parental education and performance.
 - Pass and fail percentages.
 - Grade distribution.
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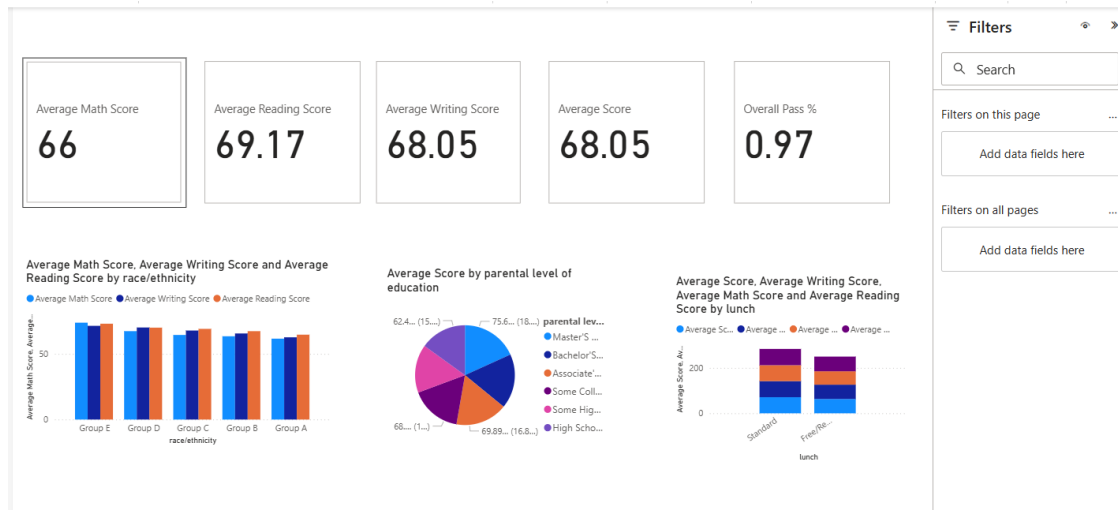
14. Result

The raw Student Performance dataset was successfully converted into a structured and analysis-ready dataset.

The final dataset contains:

Cleaned Data + Calculated Scores + Performance Grade + Pass/Fail Status + Unpivoted Subject Data + DAX Measures

This provides the required foundation for creating the Student Performance Dashboard.



15. Conclusion

The Week 7 activity helped in understanding the importance of **data cleaning and transformation in Power BI**. The raw student examination data was inspected, cleaned, transformed, and prepared for analysis.

Calculated columns and DAX measures were also created to summarize student performance. The final dataset is ready to be used for dashboard development and visualization in the next week.

Data Preparation Flow

Raw Dataset → Data Inspection → Cleaning → Transformation → Calculated Fields → Unpivot → DAX Measures → Validation → Dashboard Ready

